A Robust Approach for Brain Tumor Detection using Transfer Learning

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***Abstract*— Brain tumor is a significant health concern that requires an early diagnosis for improved treatment outcomes. Artificial Intelligence (AI) can assist to reduce the need for invasive biopsies by analyzing medical imaging data to successfully detect brain tumor. Transfer learning is an effective technique in the field of deep learning, it has developed into a promising approach for improving model performance. In this study, transfer learning model VGG16, Inception V3 and EfficientNet B2 used for brain tumor detection based on their texture, location, and shape. The Kaggle (Brain Tumor Classification - MRI) dataset was utilized for training and testing the model. The outcome of the proposed model state-of-the-art methods on result demonstrates that it is more accurate and could assist doctors make better decisions.**

Keywords— Brain Tumor Classification, Magnetic Resonance Imaging (MRI), Deep Learning, Transfer Learning, Convolutional Neural Network.

I. INTRODUCTION

Brain is the complex and essential control centre of the human body and contains around 1000 trillion synapses and 100 billion neurons. A brain tumor is an abnormal development of cells in the brain [10]. According to the report from the International Association of Cancer Registries (IARC), India records approximately 28,000 documented cases of brain tumors annually, and unfortunately, 86% of the patients die. The situation is particularly concerning in the United States [11], where primary brain tumor and CNS are expected to cause 18,990 deaths in 2023.

There are around 150 various types of brain tumors, each associated with specific affected brain tissues. Brain tumors can arise in various areas of the brain or skull, such as the protective lining, brainstem, skull base, sinuses, nasal cavity, and numerous other locations throughout the central nervous system. While not all brain tumors are cancerous, benign (noncancerous) tumors can still pose a significant risk due to their size or location. The potential dangers arise from the tumor's effects on nearby brain tissues, structures, or vital functions, rather than its cancerous nature. [1]. The seriousness of a brain tumor is determined by its grades. A pathologist will analyze the tissue obtained from the biopsy under a microscope to assess the tumor's grade. Brain tumor grading is a category system used to describe the characteristics of brain tumor cells and assess the likelihood of tumor growth and spread. The grading scale typically ranges from grade 1, which represents the least aggressive tumors, to grade 4, which indicates the most aggressive tumors.

In this study, a transfer learning model was employed to identify and classify glioma tumor, meningioma tumor, pituitary tumor, and no tumor based on their texture, position, and shape [2]. A glioma tumor arises in the spinal cord or brain. Meningioma originates in the meninges, which are the membranes that cover the spinal cord and brain. The pituitary tumor appears in the pituitary gland, a small gland located at the base of the brain [5]. Oncologists frequently use medical imaging technologies like MRI and CT scans [6] to detect and monitor brain tumor. These methods can produce a high-resolution picture of the brain tumor [8]. In this study, Kaggle MRI-collected dataset [9] of grayscale images containing of different categories of brain images is used.

This study introduces an analysis of three transfer-learning architectures that is VGG16, Inception V3 and EfficientNet B2 applied to the classification and detection of brain tumor images obtained through MRI images. VGG16 consists of 16 layers of convolutional layers, 5 layers of max-pooling layers, and 3 layers of fully connected layers. Inception V3 is a more complicated architecture that extracts information from an input image by combining parallel convolutional layers with varying filter sizes and pooling techniques [14]. EfficientNet B2 is a state-of-the-art convolutional neural network that utilizes a combination of depth, width, and resolution scaling to improve accuracy and efficiency. The result will contribute to the field of medical image analysis by demonstrating the effectiveness of transfer learning in improving brain tumor detection accuracy and efficiency. The research outcomes will have significant implications for clinical practice, enabling early detection of brain tumors and facilitating timely interventions for improved patient care.

II. LITERATURE SURVEY

The most prevalent research on the topic which is the detection of brain tumor based on deep learning and CNNs will be discussed in this literature review, along with further relevant reference works. The use of transfer learning for brain tumor identification and detection has possible to significantly improve patient diagnosis and treatment planning. The finding and categorization of brain tumor is an important challenge due to their presence.

Sushreeta Tripathya et. al [1], In order to address the constraint of limited sample size and obtain better outcomes than CNN architectures, a pre-trained EfficientNet model was built. To increase model performance, the images are scaled, cropped, and enhanced for training with EfficientNet-B2, B3, and B4. Dillip Ranjan Nayak et. al [2], Work on dense EfficientNet model with min-max normalization. Rayene Chelghoum et. al [7], A fully automatic brain tumor classification method is proposed using a CE-MRI dataset from figshare. It achieves good results with a minimal number of images by employing nine pre-trained architectures with three epochs.

Chetana Srinivas et. al [8], Classification was performed using preprocessing techniques and augmentation method to enhance accuracy and robustness of the system. We use pre-trained CNN architectures Inception-v3, ResNet50, and VGG-16 to perform on brain MRI images. For optimizing the models, hyperparameter tuning was also used. Mohamed Ait Amou et. al [10], It uses optimization approaches to set CNN hyperparameters that classify tumor classifications. Bayesian Optimization was used to determine ideal hyperparameters for the model, and its performance compared to the same dataset was used to train and fine-tune, five pre-trained CNN. Fatma Taher et. al [11], Three models were used a CNN model, transfer-learning-based models and a model trained from scratch. The BRAIN-TUMOR-net architecture combine existing pre-trained models with substantially less processing resources and out of all the models, it achieved the highest accuracy compared to the others.

Moinul Islam et. al [13], In this research, average CNN architecture, federated learning, CNN model and voting ensemble were utilized to classify brain tumors using MRI data. They compared six distinct CNN model architectures based on test accuracy, f1 score, recall, precision and model loss. Hassan Ali Khan et. al [14], In comparison to ResNet-50, VGG-16, and Inception-v3 models that use edge detection and data augmentation approaches, a small dataset is used to achieve high accuracy and uses fewer processing resources. The VGG-16 takes 456 seconds to train every epoch, ResNet-50 required 606 seconds for processing, while Inception-v3 needs 375 seconds. Hasnain Ali Shah et. al [15], To perform better than other CNN models, a fine-tuned EfficientNet-B0 model is employed combined with image enhancement techniques and data augmentation approaches to obtain high classification accuracy.

Table 1: Literature Survey

Author, Year	Dataset Used	Number of Images	Technique	Layers	Classification Type	Accuracy (%)
Sushreeta Tripathya et. al, 2023 [01]	Kaggle Dataset	3000	EffecientNet-B2, EfficientNet-B3, EffecientNet-B4	-	-	100
Dillip Ranjan Nayak et. al, 2022 [02]	Kaggle Dataset	3060	Dense Efficient Net, CNN, Fuzzy Logic	16	Multi Class	98.78%
Rayene Chelghoum et. al, 2022 [07]	Guangzhou and General Hospital, Tianjin Medical University, Nanfang Hospital,	3064	ResNet18, ResNet50, ResNet101, ResNet-Inception-v2 and SENet, VGG16, VGG19, AlexNet, GoogleNet	18-34-50-101-152, 50-101-152, 18-34-50-101-152, 16-19, 8, 22	-	98.71%
Chetana Srinivas et. al, 2022 [08]	MRI Dataset	233	VGG-16, Inception-v3, ResNet50	16	-	0.96%, 0.78%, 0.95%
Mohamed Ait Amou et. al, 2022 [10]	Figshare Dataset	3064	DenseNet201, ResNet50, InceptionV3, VGG16, VGG19	-	Multi Class	94.81%, 89.29%, 92.86%, 97.08%, 96.43%
Fatma Taher et. al, 2022 [11]	3 Different MRI Dataset	155, 1550, 5504	InceptionResNetv2, Inceptionv3, ResNet50	-		100%, 97%, 84.78%
Moinul Islam et. al 2022 [13]	MRI dataset	2309	VGG16, VGG19, Inception V3, ResNet50, DenseNet121, Inception	-	-	96.68%
Hassan Ali Khan et. al, 2020 [14]	MRI Images	253	Inception-v3, ResNet-50, VGG-16	8	Binary Class	75%, 89%, 96%

Hasnain Ali Shah et. al, 2022 [15]	Kaggle Dataset	3762	EfficientNet-B0	-	Binary Class	98.87%

III. THE PROPOSED SYSTEM

The proposed method aims to develop a brain tumor detection system using Transfer learning. It utilizing three well-known pre-trained convolutional neural network models are VGG16, Inception V3, and EfficientNet B2. We use a publicly available brain MRI dataset containing images are glioma tumor, meningioma tumor, no tumor, and pituitary tumor. The dataset was divided into two distinct subsets, one for training the model and another for testing the model's performance. Evaluate the performance of the model based on the following metrics: F1-score, recall, precision, support, sensitivity, specificity, accuracy and loss. The model will be deployed that input brain MRI image will classify as tumor or non-tumor. By implementing this technique, doctors may be able to diagnose brain tumors more quickly and accurately, which will benefit for patients.

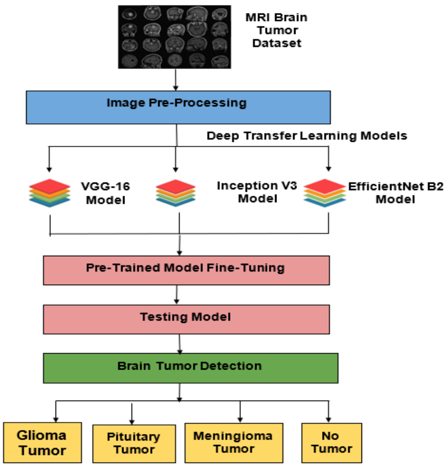

Fig. 1 Workflow Diagram of Brain Tumor Detection

3.1 Data Collection and Image Pre-processing:

We used publicly accessible MRI brain image Kaggle dataset are (https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri) [9]. The dataset utilized in this study consists of 3264 MRI brain tumor images. The dataset was split into two sections, where 80% of the data was allocated for model training, and the remaining 20% was reserved for model testing.

Table 2. Details of MRI Brain Image Dataset

Types of Brain Tumor	Images	Format	Type
Glioma Tumor	926	JPG	Grayscale
Meningioma Tumor	937		
No Tumor	500		
Pituitary Tumor	901		
Total	3264		

Table 2 displays MRI Brain Tumor Dataset images in details. For preprocessing, to ensure the consistency of the collection, images were resized to a standard dimension of 224 x 224 pixels [3]. Depending on the requirements of models, the images are either modified to grayscale [15] or RGB. Fig. 2 illustrates an MRI image from the brain tumor dataset.

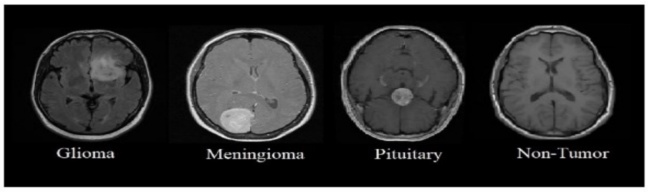

Fig. 2 Sample Images of MRI Brain Tumor Dataset

3.2 CNN Model Architecture

A Convolutional Neural Network (CNN) is a type of deep learning architecture that is commonly used to image classification, object detection, and computer vision tasks. For image detection and classification, the CNN architecture consists of a number of layers. The essential structure are Input Layer, Convolutional Layer, Pooling Layer, Fully Connected Layer, and Output Layer. The CNN architecture is depicted in Fig. 3.

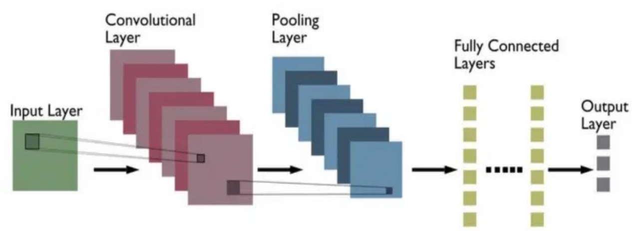

Fig. 3 Architecture of CNN

1. *Input Layer:* The image data is fed into the network through the first layer. It is often made up of an array of neurons, with each neuron representing a single pixel in the input image [10]. The width and height of the input layer are both dependent on the size of the input image.
2. *Convolution Layer:* The core layer of a CNN refers to the convolutional layer, which plays a fundamental role in extracting features from input data. A small weighted matrix for each filter is moved across the input image, computing the dot product at every location. This layer produces a group of feature maps, where correspondingly map reflects the reaction of a certain filter to the image that was fed into it.
3. *Pooling Layer:* The pooling layer main objective is to decrease the size of input data, which comes in two types: max-pooling and average-pooling. In our approach, we utilize max pooling as the pooling technique. Max pooling selects the maximum value within a defined region or receptive field of a feature map and discards the remaining values. This process results in a final feature map with reduced variables and simplified processing.
4. *Fully Connected Layer:* This layer combines and classifies the input image using the high-level information that were extracted by the previous layers.

5. *Output Layer:* The final layer is the output layer, that frequently utilises a SoftMax function to convert the preceding layer's class scores into probabilities.

3.3 Transfer Learning Approach

The purpose of detecting brain tumor on MRI images, transfer learning is an effective method. The utilized pre-trained models include VGG16, Inception V3 and EfficientNet B2. The models undergo a fine-tuned, process where they are optimized based on the desired data. They use to detect specific characteristics and patterns associated with brain tumor. This helps to prevent overfitting and allows the model to generalise well to unseen images. Once the model has been trained, it may be used to categories new MRI scans into the tumor types glioma, meningioma, pituitary, and no tumor. This can be very helpful in clinical settings, to certain conditions can be effectively treated, too fast and reliable diagnosis of brain tumor is essential. In Table 3, each transfer learning model are explained in depth.

Table 3. Transfer Learning model Parameters with values

Models	Input Size	Layers	Optimizer	Batch Size
VGG 16	224 x 224	16 Layers	Adam	32
Inception V3	299 x 299	42 Layers		
EfficientNet B2	260 x 260	23 Layers		

3.3.1 VGG-16 Model Architecture

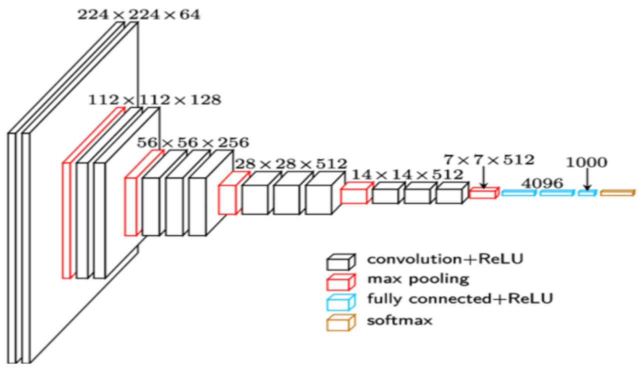

Fig. 4 VGG-16 Architecture

Fig. 4 represents the VGG-16 architecture. VGG16 is a 16-layer convolutional neural network (CNN) architecture which includes 13 convolutional layers and 3 fully connected layers [12]. The VGG16 contains 3x3 filters with stride 1, followed by ReLU activation functions. The utilization of smaller filters allows the network to capture intricate features effectively, while also reducing the overall number of parameters for the model. Following the convolutional layers are 2x2 max pooling layers with stride 2, which decreases the characteristics of these feature maps enhance the model's resilience to variations, making it more robust in the position of objects in the image. The VGG16 fully connected layers have 138 million parameters [14] and each were created for carrying out the final classification using the learned features. To create a probability distribution over the classes, The last layer of the model utilizes the SoftMax activation function.

3.3.2 Inception-V3 Model Architecture

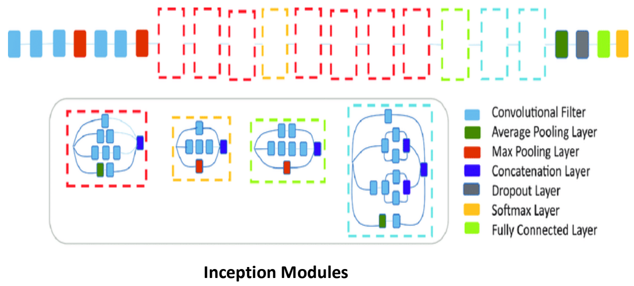

Fig. 5 Inception-V3 Architecture

Inception V3 architecture was developed by Google in 2015 specifically for image classification tasks. The model has a hierarchical structure, which allows for it to learn continuously complicated representations of the input images [15]. Fig. 5 shows the Inception V3 architecture.

A total of 42 layers forms the structure of the Inception v3 network, including various convolutional layers that have different kernel sizes, number of max pooling layers, and fully connected layers. To enhance its performance, batch normalization, dropout, and factorized convolutions were added. The model includes multiple inception modules, which are building blocks containing multiple convolutional layers that have various filter sizes. Convolutions size 1x1, 3x3, and 5x5, as well as max pooling layers, are included in these filters. The outputs of these filters are combined and subsequently fed into the next layer of the network. The auxiliary classifiers consist of a 1x1 convolutional layer followed by an average pooling layer, fully connected layer, and SoftMax activation function. These are added into the model at different stages. The 7x7 convolutional filters are factored into two distinct filters of sizes 1x7 and 7x1. As a result, the model's parameter count is decreased, and makes it more efficient.

3.3.3 EfficientNet B2 Model Architecture

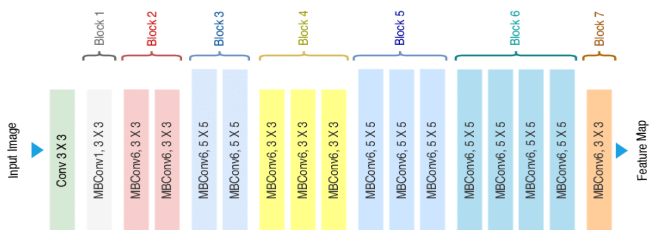

Fig. 6 EfficientNet Architecture

Fig. 6 illustrates the EfficientNet architecture. EfficientNet-B2 is a member of EfficientNet family series and represent a specific design of a convolutional neural network (CNN). By increasing the network's width, depth, and resolution in a balanced manner, the architecture is designed to improve both accuracy and efficiency [4]. The model structure made up of a stem convolutional layer that extracts feature from the input image and a number of 23 blocks that each have a set of convolutional layers, batch normalization, and activation functions. These blocks use various kernel sizes, expansion ratios and output channels to capture features in multiple scales. After that, a range of pooling layers are applied to the output of these convolutional layers, which helps compress the feature map dimensions. The features are passed through a series of entire connected layers that classify the image into different types. The output layer of the architecture consists of three nodes, representing the classes, and produces a probability distribution across these classes.

IV. RESULT ANALYSIS

In this section, the outcomes of research conducted to demonstrate the efficiency of the proposed method are explained.

4.1 Experiment Setup

Google Colab was used for the implementation. This cloud service is built on Jupyter Notebooks and offers a virtual GPU powered by an NVIDIA Tesla K80 with 12 GB of RAM. The Python model implements the TensorFlow framework using the Keras library [10].

4.2 Graph

The experiment has been executed with a batch size of 32 over 10 epochs. Graphs for VGG16, Inception V3, and EfficientNet B2 models are shown in Figures 7, 8, and 9 accordingly. In the graph, the x-axis typically represents the number of training epochs, while the y-axis represents the loss value and accuracy value. The model accuracy graph illustrates the performance of the model in terms of correctly classified samples over time. It shows in diagram the percentage of samples that are accurately classified by the model during training. The occasional fluctuations in accuracy can occur due to the complexity of the dataset or training process.

The loss diagram curve of fig. 8 should exhibit a decreasing trend, indicating that the model is learning and improving its performance. The fluctuations or sudden increases in the loss can occur in fig. 7, 9 due to various factors such as overfitting, learning rate adjustments, or changes in the training data. With an average of 97.5%, EfficientNet B2 obtained the highest accuracy, followed by InceptionV3 and VGG16 with 90.2% and 82.2% accordingly. EfficientNet B2 demonstrated the lowest model loss value with 0.13, after that InceptionV3 obtained a value 0.70 and VGG16 had the value of 0.32.

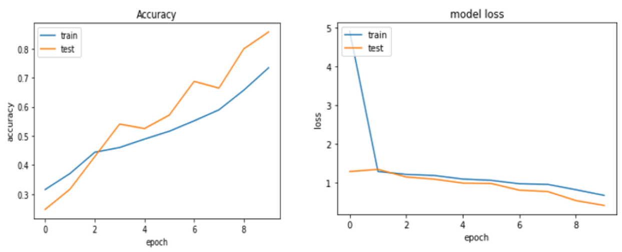

Fig. 7 Graph of Accuracy and Model Loss for VGG 16

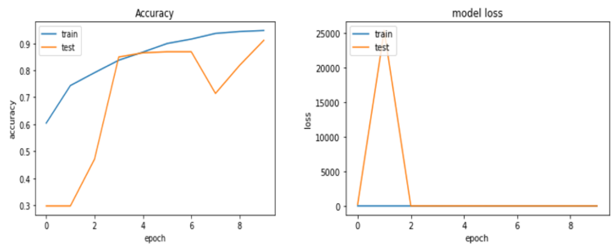

Fig. 8 Graph of Accuracy and Model Loss for Inception V3

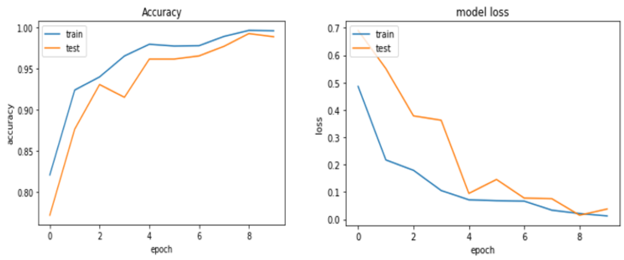

Fig. 9 Graph of Accuracy and Model Loss for EfficientNet B2

4.3 *Confusion Matrix*

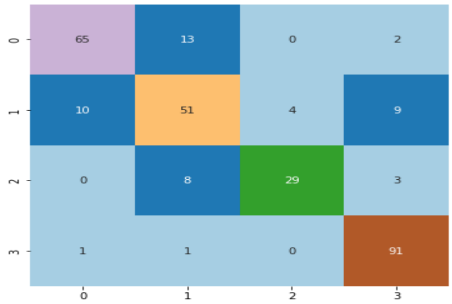

Fig. 10 Confusion Matrix VGG-16

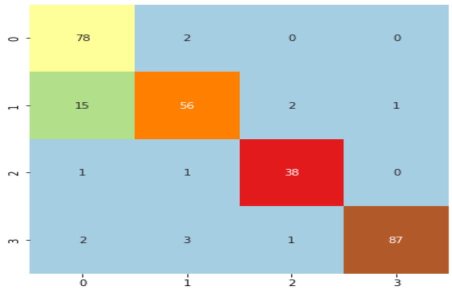

Fig. 11 Confusion Matrix Inception V3

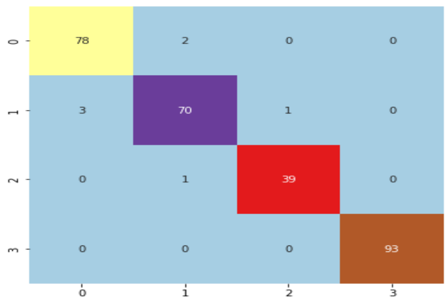

Fig. 12 Confusion Matrix EfficientNet B2

Figures 10, 11, and 12 demonstrate the confusion matrices for the VGG16, Inception V3, and EfficientNet B2 models for brain tumor detection based on glioma tumor, meningioma tumor, no tumor, and pituitary tumor. In the matrix, the diagonal cells represent the count of correctly classified samples, while the off-diagonal cells indicate the count of incorrectly classified samples. The confusion matrix gives useful information about the strengths and weaknesses of each model and can help improve the classification accuracy.

Table 4. Comparison results of the proposed model

Models	Precision%	Recall%	F1-core%	Sensitivity%	Specificity%	Accuracy%
VGG 16 Model	82.7	80.0	80.7	93.2	79.5	82.2
Inception V3 Model	90.7	90.5	90.2	96.2	90.0	90.2
Proposed EfficientNet B2 Model	**97.2**	**97.2**	**97.2**	**98.7**	**97.0**	**97.5**

Table 4, presents VGG16, Inception V3, and EfficientNet B2 optimized model comparison results based on the precision, recall, F1-score, specificity, sensitivity, accuracy. Specifically, the proposed model utilizing EfficientNet B2 obtained an overall accuracy of 97.5%. This indicates that EfficientNet B2 outperformed the other models in accurately classifying brain tumor images in the dataset.

4.4 *Comparison with State-of-the-Art Methods*

Table 5. Comparison of the accuracies of the proposed and previous state-of-the-art ML and DL methods

Related Work	Model	Accuracy%
Chetana Srinivas [8]	ResNet50	95.0
Mohamed Ait Amou [10]	DenseNet201	94.8
Fatma Taher [11]	InceptionResNetv2	96.83
Moinul Islam [13]	CNN	96.68
Proposed Method	**Efficient Net B2**	**97.5**

In order to provide an accurate evaluation of performance, the state of art model was compared to previous research. In accordance with the literature survey, the classification results comparing from the corresponding research works. Table 5 shows the comparison results in terms of accuracy. This optimized model is more accurate in classifying brain tumors. It achieved an impressive overall accuracy rate of 97.5%.

V. CONCLUSION

In this study, the proposed brain tumor detection technique using transfer learning models based on VGG16, Inception V3, and EfficientNet B2 generated excellent outcomes in accurate detection and classification based on texture, location and shape. According to the results from the study, the use of MRI brain image dataset showed that EfficientNet B2 had the most accurate accuracy of 97.5%, Inception V3 with 90.2% accuracy and VGG16 with 82.2% accuracy. This technique has the potential to significantly assist patients by enabling early detection and improved treatment.

In the future, if further developed and optimized, the model could eventually be utilized to enhance the accuracy and effectiveness of brain tumor identification. In addition, advancements in computing power and data acquisition techniques could enable the development of more powerful models, resulting in a more accurate and reliable diagnosis of brain tumor.

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